Data Collection and Analysis Using SQL

Presented by: Hemidov Ceyhun

Introduction

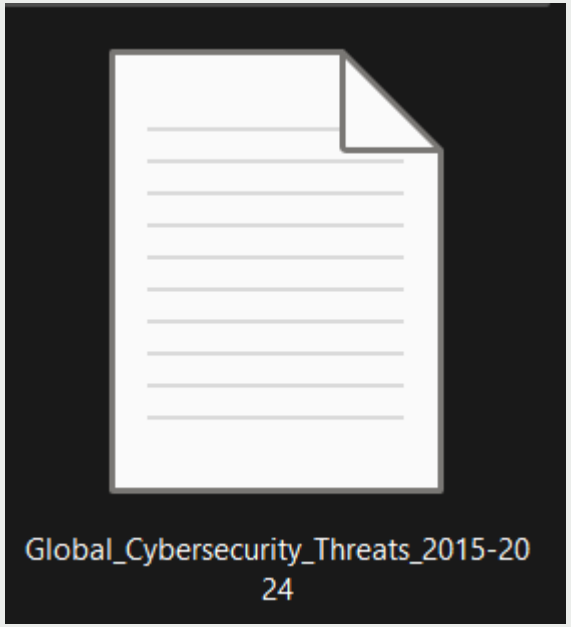
In this presentation, I will show how I collected and analyzed data using SQL.

You will learn about:

1. How I gathered the data
2. How I created data tables in SQL and imported the dataset
3. How I analyzed the data in SQL and drew meaningful conclusions

Without further delay, let’s get started!

Data Preparation

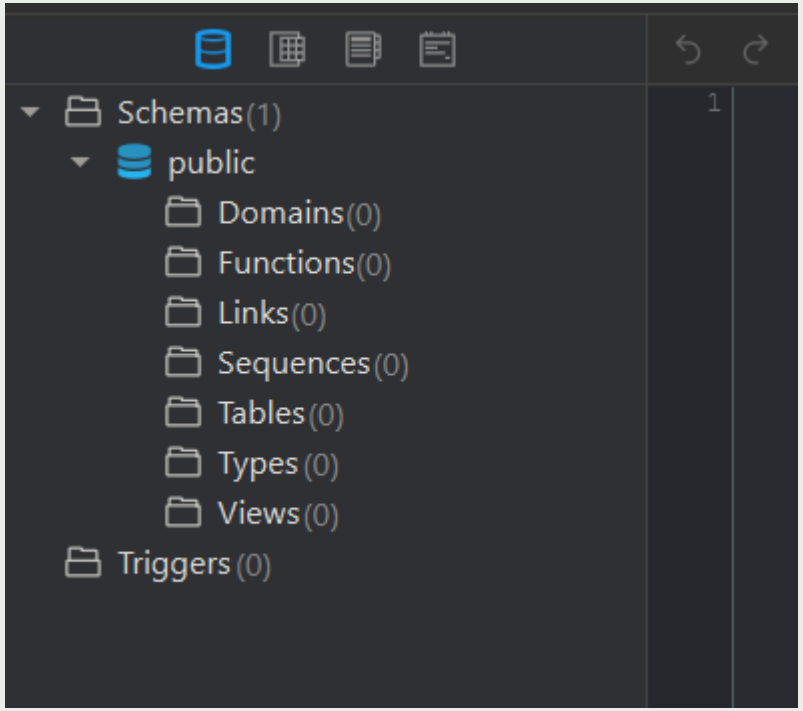


In this project, we already have a certain dataset.

However, since all of the data is in **CSV format**, it cannot be directly analyzed in SQL.

Therefore, we need to **import this data into the Valentina Studio program** to make it ready for analysis.

Creating Tables



As you can see here, there are no tables yet.

Therefore, we need to **write the SQL code for each table separately** and create them one by one.

After creating all the tables, we will be able to **perform our analyses on these tables**.

LET’S START TO CREATE DATA TABLE

Global cyber security threats

```sql
CREATE TABLE global_security_thearts(
    country VARCHAR(50),
    "year" INT,
    attack_type VARCHAR(50),
    targe_industry VARCHAR(50),
    financial_loss INT,
    number_of_affected_user INT,
    attack_source VARCHAR(50),
    security_vulnerability_type VARCHAR(50),
    defense_mechanism_used VARCHAR(50),
    incident_resolution_time INT
)
```

The columns in this dataset describe key details of each cyber incident. They include the country where the attack happened, the year of the event, and the type of cyberattack such as phishing or ransomware. The dataset also shows which industry was targeted, the financial loss in millions of dollars, and the number of affected users. Additional columns provide information about the attack source, the vulnerability that allowed the attack, and the defense mechanism used by the organization. Finally, the incident resolution time column shows how many hours were needed to fully handle the attack.

Overall, this dataset presents a clear view of global cybersecurity threats from 2015 to 2024. It highlights different attack patterns across countries, industries, and years, allowing us to identify trends and high-risk areas. The data helps analyze which attack types cause the most damage, which industries face the highest exposure, and how effectively incidents are resolved. This makes the dataset useful for security analysis, risk assessment, and developing stronger cyber defense strategies.

Now Let’s start to analyze this data

The countries that experienced the highest number of attacks.

```sql
SELECT
    country,
    round(SUM(financial_loss):: NUMERIC, 0) AS total_financial_loss
FROM global_security_thearts
GROUP BY country
ORDER BY total_financial_loss DESC
```

	country	total_financial_loss
1	UK	16503
2	Germany	15793
3	Brazil	15783
4	Australia	15403
5	Japan	15197
6	France	14972
7	USA	14812
8	Russia	14735
9	India	14566
10	China	13714

From this chart, we can see that the UK has the highest financial loss, with a total of 16,503 M dollars. It is followed by Germany, Brazil, and Australia. All 10 countries are shown here. This also shows that the difference between the country with the maximum loss and the one with the minimum loss is quite small. However, the amount of loss still reflects how strong or weak each country is in terms of cybersecurity

Countries with the highest number of cyberattacks

```sql
SELECT
    country,
    COUNT(*) AS total_attacks
FROM global_security_thearts
GROUP BY country
ORDER BY total_attacks DESC
```

	country	total_attacks
1	UK	321
2	Brazil	310
3	India	308
4	France	305
5	Japan	305
6	Australia	297
7	Russia	295
8	Germany	291
9	USA	287
10	China	281

When we compare both charts, we can see that the UK is ranked first in both categories: it has the highest number of cyberattacks and also the highest total financial loss. This shows that the UK is one of the main targets for cybercriminals. Countries like Brazil, India, France, and Japan also appear near the top in both lists, which means they face frequent attacks and suffer major financial damage. Overall, there is a clear connection: countries that experience more attacks also tend to lose more money. This suggests that the volume of attacks strongly affects the financial impact of cyber incidents.

Attack Type Distribution

```sql
SELECT
    attack_type,
    COUNT(*) AS total_attacks
FROM global_security_thearts
GROUP BY attack_type
ORDER BY total_attacks DESC
```

	attack_type	total_attacks
1	DDoS	531
2	Phishing	529
3	SQL Injection	503
4	Ransomware	493
5	Malware	485
6	Man-in-the-Middle	459

The results show that DDoS attacks are the most common, with 531 cases. Phishing is the second most frequent attack type with 529 incidents, which is almost the same as DDoS. SQL Injection and Ransomware also appear very often, showing that both application-level and data-focused attacks are major problems. Malware and Man-in-the-Middle attacks have slightly lower numbers but are still significant. Overall, the distribution shows that all six attack types are widely used, and the difference between them is not very large, meaning organizations must be ready for many different types of threats.

Industry Risk Ranking

```sql
SELECT
    target_industry AS industry,
    COUNT(*) AS total_attacks
FROM global_security_thearts
GROUP BY target_industry
ORDER BY total_attacks DESC
```

	industry	total_attacks
1	IT	478
2	Banking	445
3	Healthcare	429
4	Retail	423
5	Education	419
6	Telecommunications	403
7	Government	403

The results show that the IT sector is the most targeted industry, with 478 attacks. Banking is the second most affected sector, followed by Healthcare and Retail. Education, Telecommunications, and Government also face a high number of attacks, but their values are slightly lower. Overall, the data shows that almost all major industries are under serious cyber risk, and the difference between them is not very large. This means attackers focus on many sectors, not just one specific area.

Average Financial Loss by Industry

```sql
SELECT
    target_industry AS industry,
    round(avg (financial_loss):: NUMERIC, 0) AS
avg_financial_loss
FROM global_security_thearts
GROUP BY target_industry
ORDER BY avg financial loss DESC
```

When we compare the two charts, we see that IT, Banking, and Healthcare get the highest number of attacks. However, the Government sector has the highest average financial loss, even though it has fewer attacks. This means attacks on Government systems are less common but more damaging. On the other hand, sectors like IT and Banking face many attacks but lose slightly less money per incident. Overall, some industries are attacked more often, while others suffer more expensive attacks.

	industry	avg_financial_loss
1	Government	53
2	IT	52
3	Banking	51
4	Telecommunications	51
5	Retail	50
6	Healthcare	49
7	Education	48

Average Financial Loss by Attack Type

```
SELECT
    attack_type,
    Round(avg(financial_loss)::NUMERIC, 0) AS
avg_financial_loss
FROM global_security_thearts
GROUP BY attack_type
ORDER BY avg_financial_loss DESC
```

	attack_type	avg_financial_loss
1	DDoS	52
2	Man-in-the-Middle	51
3	Phishing	50
4	SQL Injection	50
5	Ransomware	50
6	Malware	49

The results show that DDoS attacks cause the highest average financial loss with 52 million dollars. Man-in-the-Middle attacks are the second most expensive with 51 million. Phishing, SQL Injection, and Ransomware all have an average loss of about 50 million dollars, while Malware has the lowest loss at 49 million. Overall, the differences between attack types are very small, which means all cyberattacks can cause serious financial damage.

Average Resolution Time by Country

```sql
SELECT
    country,
    round(avg(incident_resolution_time), 2) AS
avg_resolution_time
FROM global_security_thearts
GROUP BY country
ORDER BY avg_resolution_time ASC
```

	country	avg_resolution_time
1	USA	35.31
2	Russia	35.54
3	India	35.93
4	Germany	36.02
5	France	36.22
6	Australia	36.48
7	UK	36.93
8	Japan	37.1
9	China	37.4
10	Brazil	37.76

The results show that the USA, Russia, and India have the fastest resolution times, which means they are the most efficient at handling cyber incidents. Germany, France, and Australia are in the middle range, while the UK, Japan, China, and Brazil take longer to resolve attacks. Overall, countries with faster resolution times likely have stronger security teams and better response systems, while slower countries may need improvements in their incident management process.

Vulnerability Type Frequency

```sql
SELECT
    security_vulnerability_type AS vulnerability,
    count(*) AS total_occurrences
FROM global_security_thearts
GROUP BY security_vulnerability_type
ORDER BY total_occurrences DESC
```

The results show that Zero-day vulnerabilities are the most common, appearing 785 times. Social Engineering is the second most frequent, followed by Unpatched Software and Weak Passwords. This means attackers often take advantage of unknown security flaws and human mistakes. Technical weaknesses like missing updates or poor passwords are also major risk factors. Overall, all four vulnerability types are widely used by attackers, and organizations must address both human and technical weaknesses to reduce cyber risks.

	vulnerability	total_occurrences
1	Zero-day	785
2	Social Engineering	747
3	Unpatched Software	738
4	Weak Passwords	730

Defense Mechanism Effectiveness

```sql
SELECT
    defense_mechanism_used AS defense,
    count(*) AS total_attacks
FROM global_security_thearts
GROUP BY defense_mechanism_used
ORDER BY total_attacks DESC
```

	defense	total_attacks
1	Antivirus	628
2	VPN	612
3	Encryption	592
4	Firewall	585
5	AI-based Detection	583

The results show that attacks are most successful against systems that use Antivirus and VPN protection, with 628 and 612 incidents. Encryption and Firewall also face a high number of successful attacks, while AI-based Detection has the lowest number but still remains significant. This means that traditional security tools like Antivirus, VPN, and Firewall are not always strong enough to stop modern cyberattacks. Even advanced systems like AI detection cannot fully prevent them. Overall, all defense mechanisms can be bypassed if attackers use more advanced techniques.

Year-on-Year Attack Trends

```sql
SELECT
    YEAR,
    Count(*) AS total_attacks
FROM global_security_thearts
GROUP BY YEAR
ORDER BY total_attacks DESC
```

	year	total_attacks
1	2017	319
2	2022	318
3	2020	315
4	2023	315
5	2018	310
6	2021	299
7	2024	299
8	2016	285
9	2015	277
10	2019	263

The results show that cyberattacks do not follow a simple increasing or decreasing pattern. From 2015 to 2017, the number of attacks rises steadily and reaches its highest point in 2017. After that, the numbers stay mostly stable, with small ups and downs between 2018 and 2024. The lowest value is in 2019, while 2020–2023 remain in a similar range. Overall, the trend is fairly stable with slight fluctuations, meaning cyber threats remain consistently high each year.

Overall Conclusion

Overall, our analysis shows that cyberattacks are a serious problem for almost all countries and industries. The UK is the most affected country in both attack count and financial loss. IT and Banking face the highest number of attacks, but Government has the highest average loss per incident, which was surprising. We also saw that different attack types cause almost the same amount of damage, and there is almost no connection between affected users and financial loss. Some countries, like the USA and Russia, solve incidents faster, while others need more time. Common weaknesses such as Zero-day flaws and Social Engineering appear very often, and even strong defense tools like Antivirus or VPN are not always enough. This shows that organizations must improve both technical security and response skills.

Thank you for your attention!